

README

UserManagementAPI Project

Overview

UserManagementAPI is an ASP.NET Core Web API project developed in **Visual Studio 2026**. It provides full CRUD functionality for managing user records and includes middleware for security headers, input sanitization, rate limiting, and global exception handling. This guide explains how peer reviewers can download, open, build, run, and test the project successfully. It includes instructions for both Visual Studio and VS Code users.

1. Prerequisites

Before running the project, install the following software:

Required Software

- **.NET 8 SDK:** Download it from <https://dotnet.microsoft.com/en-us/download/dotnet/8.0>.
- **Git** <https://git-scm.com/downloads>
- **IDE options:** Use Visual Studio 2022/2026 or VS Code.
 - **Visual Studio 2022/2026** (recommended)
 - Install the *ASP.NET and Web Development* workload
 - **VS Code**
 - Install the **C# Dev Kit** extension
 - Install the **.NET Install Tool** extension (optional but helpful)

2. Download the Project from GitHub

Option A: Git Command Line

Open a terminal and run the following command:

```
git clone https://github.com/<your-repo-name>/UserManagementAPI.git
```

Option B: GitHub Desktop

1. Open GitHub Desktop.
2. Click **Clone Repository**.
3. Select the UserManagementAPI repository.
4. Choose the local folder where you want to save the project.

3. Open the Project

VS Code Instructions

1. Open VS Code.
2. Select **File > Open Folder**.
3. Choose the cloned project folder.
4. VS Code should detect the `.csproj` file.
5. When prompted, click **Restore** to install the project dependencies.

Visual Studio 2026 Instructions

1. Open Visual Studio 2026.
2. Select **Open a project or solution**.
3. Choose the project's `.csproj` file.
4. Visual Studio automatically restores the required NuGet packages.

4. Run the API

Using VS Code

Open the integrated terminal and run these commands:

```
bash
dotnet build
dotnet run
```

After the API starts, you should see a message like this:

Now listening on: <https://localhost:5001>

Using Visual Studio 2026

Press **F5** or click **Start Debugging**.

Visual Studio will automatically:

- Build the project.
- Launch the API.
- Open Swagger in your browser.

5. Test the API with Swagger

After the API is running, test it in Swagger by following these steps:

1. Open your browser.
2. Go to <https://localhost:5001/swagger/index.html>.
3. Review the available API endpoints.
 - **Users** endpoints
 - (Auth endpoints appear in Phase 3 versions)
4. Expand an endpoint, click **Try It Out**, enter the required JSON, and then click **Execute**.

6. Common Issues and Solutions

HTTPS Certificate Warning in VS Code

If VS Code prompts you to trust the development certificate, run this command:

```
dotnet dev-certs https --trust
```

Port Already in Use

If the default port is already in use, run the API on a different port:

```
dotnet run --urls https://localhost:6001
```

Missing Dependencies

If dependencies are missing, restore the project packages:

```
dotnet restore
```

SQLite Database Not Created

The API automatically creates the SQLite database file the first time it runs. If the database file is deleted, run the API again to recreate it.

7. Notes for VS Code Users

VS Code does **not** automatically perform the following tasks:

- Generate launch profiles.
- Attach debuggers automatically.
- Open Swagger automatically.

To run the project in VS Code, start the API manually:

```
dotnet run
```

Then open Swagger manually in your browser. This behavior is normal and expected when using VS Code.

8. Notes for Visual Studio 2026 Users

Visual Studio provides the following built-in conveniences:

- Automatic project builds.
- Automatic API launch.
- Integrated debugging tools.
- Automatic Swagger launch.

These features help the project run more smoothly in Visual Studio 2026.

<https://github.com/MrGDawg/Back-End-API-Project.git>